DP Biology – D1.3 (HL extension)

Editing and switching off genes

- How do mutations in genes arise?
- What effects do gene mutations have?

HL statements D1.3.8–10. Textbook pp. 643–647.



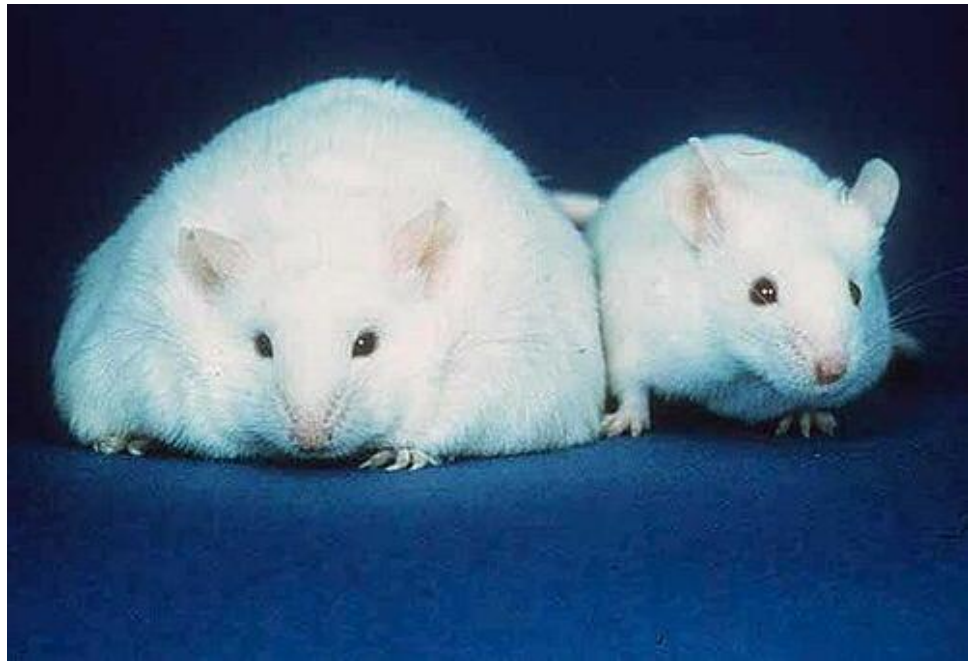
Image: a mouse lacking a gene that controls appetite (left) beside a normal mouse. Public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	D1.3.8–9	Explain how gene knockout reveals a gene's function, and how CRISPR-Cas9 edits genes
2	D1.3.10	Evaluate two hypotheses for why some gene sequences are highly conserved

Gene knockout

D1.3.8



Knockout mouse (left) and normal mouse. Public domain.

A chosen gene is made **inoperative**, usually by deleting or disrupting part of it.

The knockout's phenotype is compared with normal animals: the difference shows what the gene does.

Example

Mice lacking the gene for the hormone leptin overeat and become obese, so leptin helps control appetite.

The logic of a knockout

D1.3.8



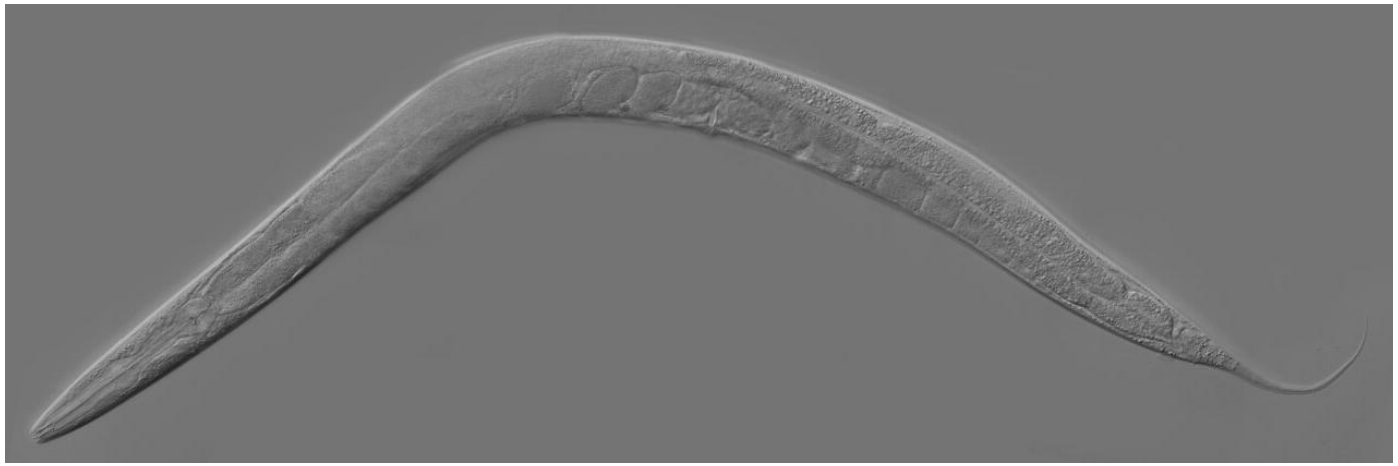
Gene knocked out	Phenotype seen	Conclusion
p53	Many tumours	Helps prevent cancer
Leptin	Overeating, obesity	Signals fullness

Libraries

Collections of knockout strains exist for mice and other model species; researchers order the one they need.

Why model organisms work

D1.3.8



The nematode Caenorhabditis elegans. CC BY-SA 2.5.

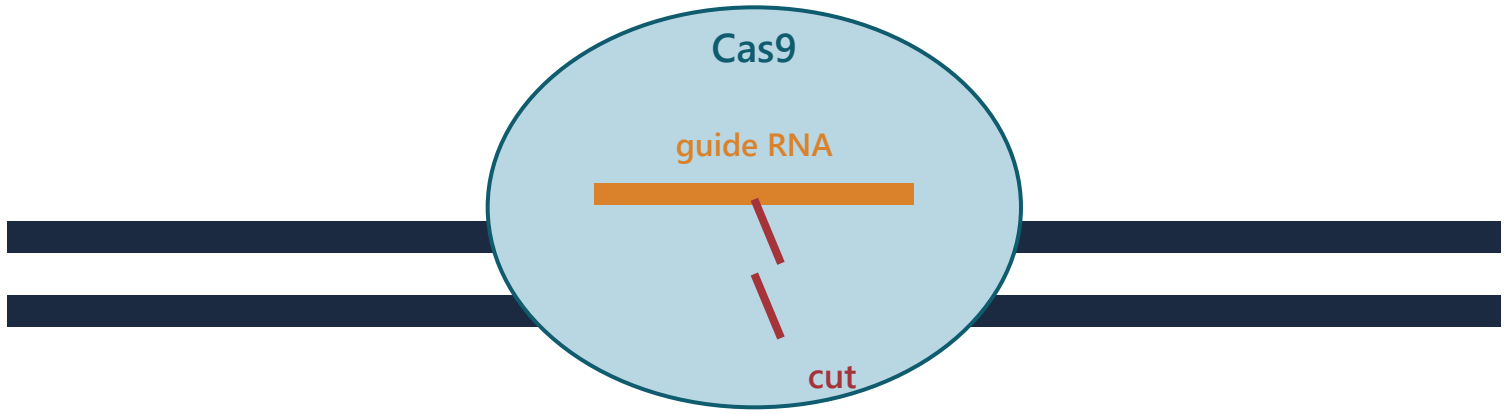
Organism	Useful because
Mouse	A mammal, close to humans
Fruit fly	Short life cycle, many offspring
C. elegans	Transparent; every cell mapped

Evaluate

Shared genetic code and many shared genes mean results can apply to humans. But a gene may have different roles in different species, and another gene may cover for the missing one. What about the welfare of the animals?

How CRISPR-Cas9 cuts DNA

D1.3.9

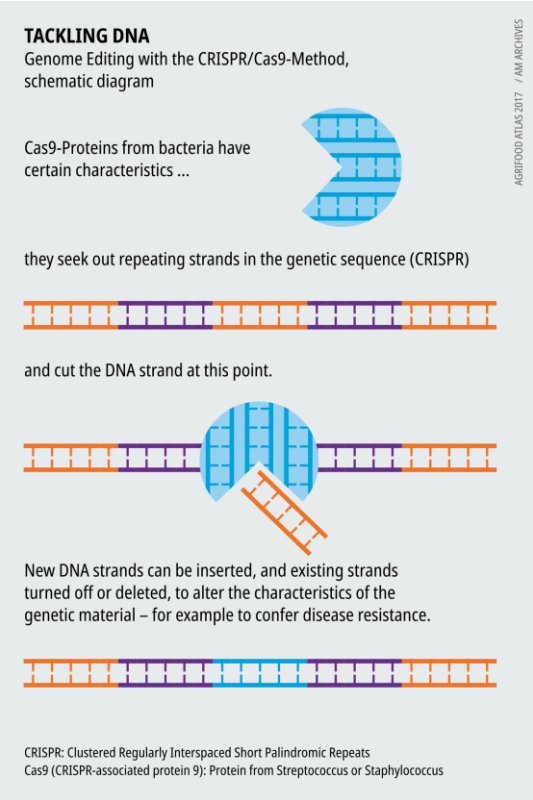


A **guide RNA** is designed to match the target sequence. It leads the enzyme **Cas9** there.

Cas9 cuts **both strands**. When the cell repairs the cut, the gene can be disabled or new DNA inserted.

Where CRISPR came from

D1.3.9



Genome editing with CRISPR-Cas9. CC BY-SA 4.0.

CRISPR and Cas9 are a **bacterial defence** against viruses: the bacterium stores pieces of viral DNA and cuts matching DNA if the virus returns.

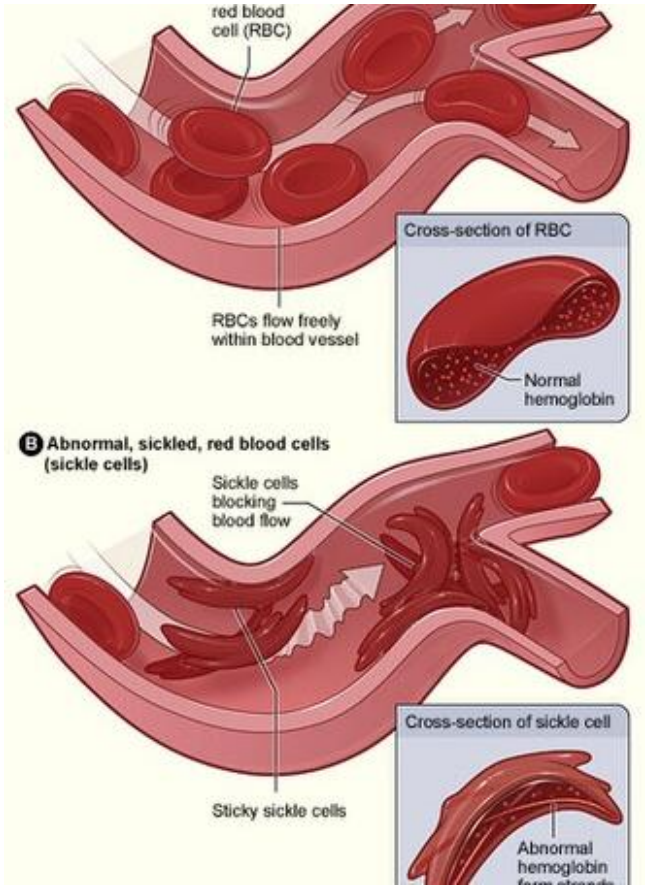
Scientists change the guide RNA to send Cas9 to almost any sequence.

Why it matters

Cheaper, faster and more precise than older methods: used in research, crops and medicine.

A successful use: sickle cell disease

D1.3.9



Normal and sickle red blood cells. Public domain.

Blood stem cells are taken, edited with CRISPR-Cas9 to switch **foetal haemoglobin** back on, and returned.

Foetal haemoglobin does not sickle. The treatment was approved in 2023.

Discuss

It is very expensive and needs strong chemotherapy first. Who should get access?

Editing step by step

D1.3.9



Ethics

Editing body cells affects only the patient. Editing embryos would pass changes to future generations. Which uses are acceptable, and who decides?

Practice: design the experiment

D1.3.8–9

Unknown gene

A gene is active only in mouse kidneys. How could you find its role?

Guide RNA

Write a guide RNA for the DNA target TACGGA (pairs with this strand).

Off target

Suggest a risk if the guide matches two places in the genome.

Explain

Give a reason for each answer.

Working with data: a knockout study

D1.3.8

Mean body mass of mice at 12 weeks. The values are invented for practice; the pattern matches published studies.

Mice	Body mass / g
Normal	28
Leptin gene knocked out	62
Knockout given leptin injections	31

1. Calculate the percentage increase in mass of knockouts.
2. Explain the result for the injected mice.
3. Explain why the injected group strengthens the conclusion.

Practice: true or false?

Bacteria

CRISPR evolved to edit human genes.

Both strands

Cas9 cuts both DNA strands.

Inherited

Editing a patient's blood stem cells changes their children's genes.

For each one

Say true or false and correct any false statement.

Check-in: class 1

D1.3.8–9

1. Define gene knockout.
2. Explain how a knockout shows a gene's function.
3. Give one limitation of using model organisms.
4. Outline how CRISPR-Cas9 edits a gene.
5. Describe one medical use of CRISPR.

How to use this

Answer without your notes, then check against slides 3–12.

Conserved sequences

D1.3.10

Conserved sequences are the same or similar across species. **Highly conserved** ones have barely changed over vast spans of evolution: histone and ribosomal RNA genes are shared by yeast and humans.

Hypothesis	Explanation
Functional requirement	The product only works with its exact sequence; changes are harmful and removed by selection
Slower mutation rate	These regions mutate less, or are repaired better, so fewer changes arise

Evidence

Silent sites change faster than sites that alter the amino acid: support for the first hypothesis.

Spot the conserved region

D1.3.10

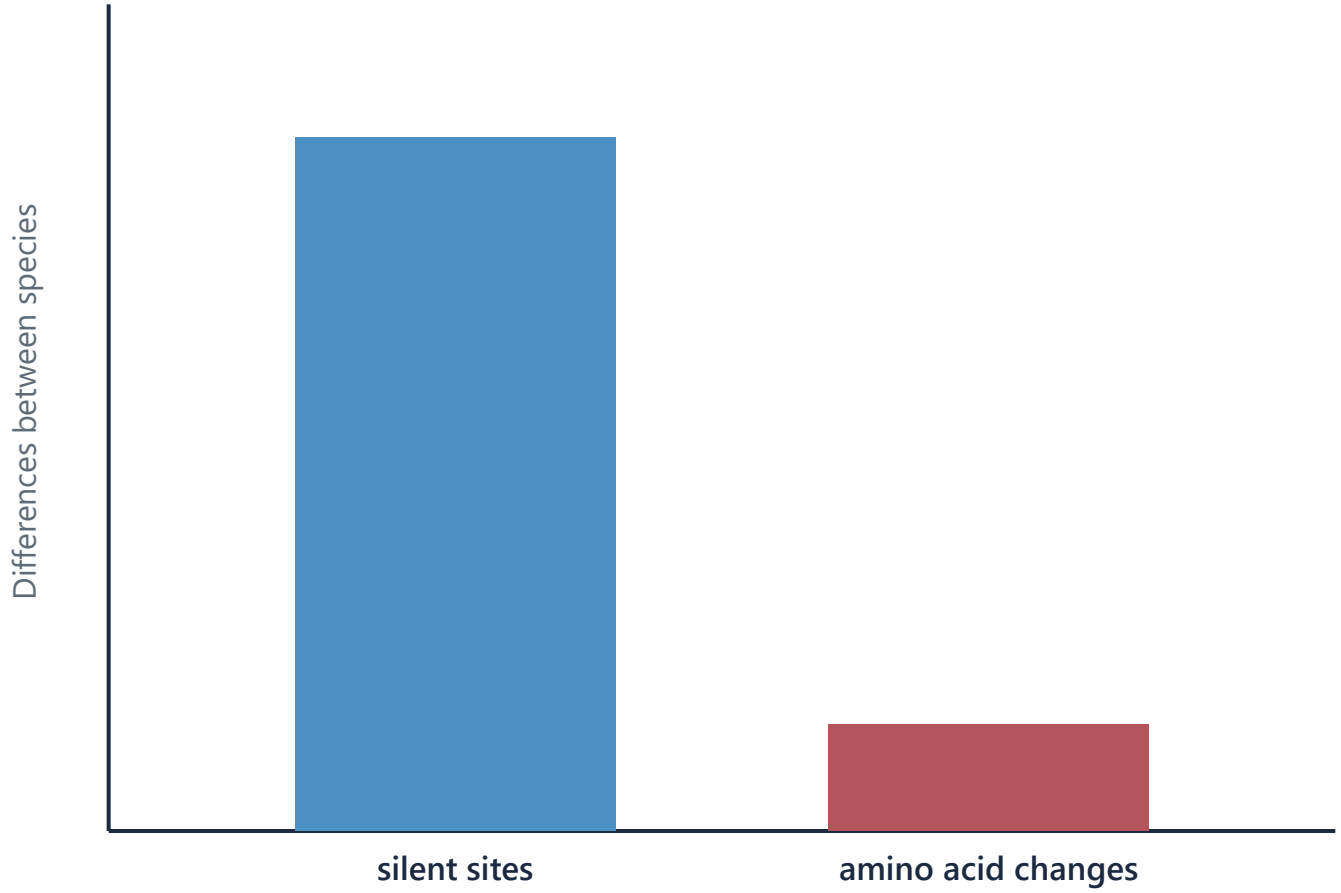
Species	First 20 amino acids of histone H4
Human	S G R G K G G K G L G K G G A K R H R K
Cow	S G R G K G G K G L G K G G A K R H R K
Fruit fly	S G R G K G G K G L G K G G A K R H R K
Pea	S G R G K G G K G L G K G G A K R H R K

Activity

How many differences can you find? Pea and human last shared an ancestor over a billion years ago. Which hypothesis best explains this?

Silent and amino-acid-changing sites

D1.3.10



Drawn sketch of a typical pattern.

In a conserved gene, **silent** changes build up between species, but changes to the **amino acids** are rare.

Mutations happen at both, so selection must be removing the harmful ones.

Working with data: a conserved protein

D1.3.10

Amino acid differences in cytochrome c compared with humans (published values).

Species	Differences
Chimpanzee	0
Rhesus monkey	1
Dog	11
Chicken	13
Yeast	45

1. Identify the closest relative of humans.
2. Explain why cytochrome c can be compared across such different organisms.
3. Suggest why differences rise with evolutionary distance.

Not all genes are conserved

D1.3.10

Highly conserved

Histones, rRNA, cytochrome c: changes break vital functions.

Highly variable

Immune system genes: variety helps fight changing pathogens.

Non-coding

Many changes have no effect, so they accumulate freely.

Think

Why might selection favour variation in genes for immune proteins but not in histone genes?

Practice: which hypothesis?

D1.3.10

Observation 1

Silent sites in a gene differ widely between mammals.

Observation 2

The same region is copied with unusually accurate repair.

Observation 3

A tiny change to a ribosomal RNA kills the cell.

For each one

Which hypothesis does it support: functional requirement or slower mutation? Explain.

Practice: compare two genes

D1.3.10

Changes per 100 sites between human and mouse. The values are invented for practice; the pattern is realistic.

Gene	Silent sites	Amino acid sites
Histone H3	30	0.2
Immune receptor	32	18

1. Compare the two genes.
2. Do the data support a slower mutation rate for histones?
3. Explain the immune receptor result.

Check-in: class 2

D1.3.10

1. Define a conserved sequence.
2. Give two examples of highly conserved genes.
3. State the two hypotheses for conservation.
4. Describe evidence that supports the functional requirement hypothesis.
5. Suggest why some genes are highly variable.

How to use this

Answer without your notes, then check against slides 14–20.

Common mistakes to avoid

Not this	But this
CRISPR adds genes from viruses.	CRISPR uses a guide RNA and Cas9 to cut a chosen sequence.
A knockout removes a whole chromosome.	A knockout inactivates one gene.
Conserved means the gene is not used.	Conserved genes are often essential.

Summary: editing and conserved genes

D1.3.8–10

	Key idea
D1.3.8	Knockouts reveal function; model organisms and their limits
D1.3.9	Guide RNA and Cas9 cut a chosen site; sickle cell treatment
D1.3.10	Conserved sequences: functional requirement or slower mutation

Check that you can...

1. **Outline** gene knockout as a way to investigate gene function
2. **Describe** how CRISPR-Cas9 edits genes
3. **Discuss** the uses and ethics of gene editing
4. **Evaluate** hypotheses for conserved sequences

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State what a gene knockout is.
2. Outline the role of the guide RNA in CRISPR-Cas9 editing.
3. Give one hypothesis for why some sequences are highly conserved.

Image credits

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Adult Caenorhabditis elegans.jpg – The original uploader was Kbradnam at English Wikipedia. (Or – CC BY-SA 2.5

Genome Editing with the CRISPR-Cas9-Method, schematic diagram.svg – Bartz/Stockmar – Agrifood Atlas, 2017 – CC BY-SA 4.0

Sickle cell 01.jpg – The National Heart, Lung, and Blood Institute (NHLBI) – Public domain

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